

Note: This document contains the full text of the survey students saw during the follow-up their senior year. We have highlighted in green the items that were analyzed for the present study.

Thank you for your help with this study!

The goal of this survey is to learn more about students' experiences in [Intro Psych] and at [School] so we can improve [Intro Psych] and courses like it for future students. We will ask questions about your experiences at [School], your beliefs, and your memory of what you learned in [Intro Psych]. Because we really care about this, we'll give you a \$20 Amazon gift card for your assistance. The survey should take about 20-30 minutes.

Note: This study is by invitation-only. If you did not receive an email invitation from the research team, you are not eligible to participate.

Important Note In order to get your honest, thoughtful reflections, we ask that you complete this survey on your own, without interruption, in a quiet location. It's important that you share your honest opinion.

Reminder:

- Your participation is voluntary.
- You may choose not to answer specific questions. Your responses will be kept completely confidential, so please share your candid answers to the questions.
- If you have any questions or concerns, please contact Bridgette Hard at [email] or Shannon Brady at email.

On the next page, you will see the study consent form. Please read it and then begin the survey.

[School] University Research Information Form
Protocol Directors: Bridgette Hard & Shannon Brady Protocol
Title: Academic Experiences – [Intro Psych]
IRB Approval date: 12/1/16
Expiration date: 11/30/19

DESCRIPTION: You are invited to participate in a research study on your academic experiences at [School] and with [Intro Psych], as well as your beliefs and attitudes about topics relevant to psychology. In the study, you will be asked multiple-choice and open-ended questions about your experiences, knowledge, and beliefs. You will also be asked to release your education record (e.g., transcript) from [School] for the purposes of the study. (You may decline this part of the study without affecting your participation in the other parts of the study.) Responses you provide in this survey may be connected with data from when you took [Intro Psych] (e.g., grades, survey responses) so that we can better understand students' experiences.

TIME INVOLVEMENT: Your participation in this study should take approximately 30 minutes.

RISKS AND BENEFITS: There are no foreseeable risks associated with this study. We cannot and do not guarantee or promise that you will receive any benefits from this study. However, you may benefit from the knowledge that you are helping [School] better understand students' experiences with [Intro Psych] and in general. The research team takes your privacy seriously and will take strong precautions to protect the privacy of your identifiable information, including the use of de-identification protocols prior to analysis and encrypted data storage systems. The research team will do our best to protect your confidentiality, though we cannot guarantee full protection against unforeseen events outside the research team's control. Your decision to participate in this study or not will not affect your grades or other aspects of your relationship with [School] University.

PAYMENTS: You will receive a \$20 Amazon gift card for participating in this study.

PARTICIPANT'S RIGHTS: If you have read this form and have decided to participate, please understand your participation is voluntary and you have the right to withdraw your consent or discontinue participation at any time without penalty or loss of benefits to which you are otherwise entitled. The alternative is not to participate. You have the right to not answer particular questions. The results of this research may be presented in de-identified and/or aggregated form at scientific or professional meetings or published in scientific journals. Your identity and your identifiable information will not be made known for these purposes.

CONTACT INFORMATION: Questions: If you have any questions, concerns or complaints about this research, its procedures, risks and benefits, contact the Co-Protocol Director, Bridgette Hard, at [email]. Independent Contact: If you are not satisfied with how this study is being conducted, or if you have any concerns, complaints, or general questions about the research or your rights as a participant, please contact the [School] Institutional Review Board

(IRB) to speak to someone independent of the research team at [phone_number] or toll free at [phone_number]. You can also write to the [School] IRB, [School] University, [address].

You may print a copy of this form to keep.

By continuing to the next page, you indicate your agreement to participate in this research.

The first questions are about your experience in general at [School].

For each question or statement, mark or give the response that is true or best for you. There are no right or wrong answers. Please use the whole range of each scale.

All things considered, on the whole how positive or negative has your experience been at [School]?

- ☐ Extremelynegative1
- ☐ Very negative2
- ☐ Quite negative 3
- ☐ Somewhat negative 4
- ☐ Slightly negative 5
- ☐ Slightly positive 6
- ☐ Somewhat positive 7
- ☐ Quite positive 8
- ☐ Very positive 9
- ☐ Extremelypositive10

I feel like I belong at [School].

- ☐ Strongly disagree
- ☐ Disagree
- ☐ Somewhat disagree
- ☐ Neither agree nor disagree
- ☐ Somewhat agree
- ☐ Agree
- ☐ Strongly agree

I fit in well at [School].

- ☐ Strongly disagree
- ☐ Disagree
- ☐ Somewhat disagree
- ☐ Neither agree nor disagree
- ☐ Somewhat agree
- ☐ Agree
- ☐ Strongly agree

I feel like an outsider at [School].

- ☐ Strongly disagree
- ☐ Disagree
- ☐ Somewhat disagree
- ☐ Neither agree nor disagree
- ☐ Somewhat agree
- ☐ Agree
- ☐ Strongly agree

I am similar to the kind of people who succeed at [School].

- ☐ Strongly disagree
- ☐ Disagree
- ☐ Somewhat disagree
- ☐ Neither agree nor disagree
- ☐ Somewhat agree
- ☐ Agree
- ☐ Strongly agree

It is a mystery to me how [School] works.

- ☐ Strongly disagree
- ☐ Disagree
- ☐ Somewhat disagree
- ☐ Neither agree nor disagree
- ☐ Somewhat agree
- ☐ Agree
- ☐ Strongly agree

When you think about [School], how often, if ever, do you wonder: "Maybe I don't belong here"?

- ☐ Always
- ☐ Frequently
- ☐ Sometimes
- ☐ Hardly ever
- ☐ Never

On a day-to-day basis, how stressful has your life been at [School]?

- ☐ Not at all stressful1
- ☐ 2
- ☐ 3
- ☐ 4
- ☐ 5
- ☐ 6
- ☐ 7
- ☐ 8
- ☐ 9
- ☐ Extremely stressful10

In the last month of school, how often have you felt that you were unable to control the important things in your life?

- ☐ Never
- ☐ Almost never
- ☐ Sometimes
- ☐ Fairly often
- ☐ Very often

In the last month of school, how often have you felt confident about your ability to handle your personal problems?

- ☐ Never
- ☐ Almost never
- ☐ Sometimes
- ☐ Fairly often
- ☐ Very often

In the last month of school, how often have you felt that things were going your way?

- ☐ Never
- ☐ Almost never
- ☐ Sometimes
- ☐ Fairly often
- ☐ Very often

In the last month of school, how often have you felt difficulties were piling up so high that you could not overcome them?

- ☐ Never
- ☐ Almost never
- ☐ Sometimes
- ☐ Fairly often
- ☐ Very often

The next three questions are about how you think about stress in general. To what extent do you agree or disagree with each of the following statements regarding stress?

Experiencing stress enhances performance and productivity.

- ☐ Strongly disagree
- ☐ Somewhat disagree
- ☐ Neither agree nor disagree
- ☐ Somewhat agree
- ☐ Strongly agree

Experiencing stress inhibits learning and growth.

- ☐ Strongly disagree
- ☐ Somewhat disagree
- ☐ Neither agree nor disagree
- ☐ Somewhat agree
- ☐ Strongly agree

The effects of stress are positive and should be utilized.

- ☐ Strongly disagree
- ☐ Somewhat disagree
- ☐ Neither agree nor disagree
- ☐ Somewhat agree
- ☐ Strongly agree

Over the past few years, we have made some changes to the exams and assignments in [Intro Psych]. The next questions will ask about your experiences with exams and assignments at [School].

Please rate your agreement with each of the following statements:

When working on a course project, I think about the overall goal of the project and what I am supposed to learn from it.

- ☐ Strongly disagree
- ☐ Disagree
- ☐ Somewhat disagree
- ☐ Neither agree nor disagree
- ☐ Somewhat agree
- ☐ Agree
- ☐ Strongly agree

When working a course project, I try to break the project into smaller, more manageable steps.

- ☐ Strongly disagree
- ☐ Disagree
- ☐ Somewhat disagree
- ☐ Neither agree nor disagree
- ☐ Somewhat agree
- ☐ Agree
- ☐ Strongly agree

When working on a course project, I do a little bit at a time, rather than putting off tasks to the last minute.

- ☐ Strongly disagree
- ☐ Disagree
- ☐ Somewhat disagree
- ☐ Neither agree nor disagree
- ☐ Somewhat agree
- ☐ Agree
- ☐ Strongly agree

When working on a course project, I feel comfortable reaching out to my instructor or TA for support and feedback.

- ☐ Strongly disagree
- ☐ Disagree
- ☐ Somewhat disagree
- ☐ Neither agree nor disagree
- ☐ Somewhat agree
- ☐ Agree
- ☐ Strongly agree

In general, when you take exams at [School], how anxious do you feel?

- ☐ 1 = Not at all anxious
- ☐ 2
- ☐ 3
- ☐ 4
- ☐ 5
- ☐ 6
- ☐ 7 = Extremely anxious

In general, when you take exams at [School], how worried about being anxious do you feel?

- ☐ 1 = Not at all worried
- ☐ 2
- ☐ 3
- ☐ 4
- ☐ 5
- ☐ 6
- ☐ 7 = Extremely worried

In general, when you take exams at [School], how confident do you feel?

- ☐ 1 = Not at all confident
- ☐ 2
- ☐ 3
- ☐ 4
- ☐ 5
- ☐ 6
- ☐ 7 = Extremely confident

When you feel anxious before an exam, what do you do?

How much do you use each of the following study strategies before exams at [School]?

	Never	Sometimes	Most of the time	Always
Memorizing definitions	☒	☐	☐	☐
Re-reading the textbook	☐	☐	☐	☐
Reviewing lecture slides	☐	☐	☐	☐
Reviewing my own notes	☐	☐	☐	☐
Rewriting notes	☐	☐	☐	☐
Creating mnemonics	☐	☐	☐	☐
Studying with friends	☐	☐	☐	☐
Quizzing myself	☐	☐	☐	☐
Reviewing prior assignments or problem sets	☐	☐	☐	☐
Attending office hours to ask about confusing material	☐	☐	☐	☐

How much you do each of the following while studying for exams at [School]?

	Never	Sometimes	Most of the time	Always
Listen to music	☐	☐	☐	☐
Have television or streaming video on in the background	☐	☐	☐	☐
Work near people who are not studying (e.g., roommates, friends, or family)	☐	☐	☐	☐
Respond to emails or text messages	☐	☐	☐	☐

We are interested in the different ways people respond when they feel anxious before an exam. For you, how effective do you think each of these strategies would be if you used it to respond to test anxiety shortly before or during an exam?

	Not effective at all	Not very effective	Somewhat effective	Quite effective	Very effective
Try to relax and calm yourself down	☐	☐	☐	☐	☐
Tell yourself that you are excited instead of anxious	☐	☐	☐	☐	☐
Remind yourself that the physiological response associated with anxiety (e.g., increased adrenaline, “feeling pumped up”) could actually help you do well, not worse	☐	☐	☐	☐	☐
Do something to take your mind off the test	☐	☐	☐	☐	☐
Ignore your anxiety: Focus on studying or focus on the test instead	☐	☐	☐	☐	☐
Spend a few moments writing about your worries	☐	☐	☐	☐	☐

Regardless of whether you personally use the strategy, how familiar are you with each of the following strategies as a way to respond to test anxiety?

	Not familiar at all	Not very familiar	Somewhat familiar	Quite familiar	Very familiar
Try to relax and calm yourself down	☐	☐	☐	☐	☐
Tell yourself that you are excited instead of anxious	☐	☐	☐	☐	☐
Remind yourself that the physiological response associated with anxiety (e.g., increased adrenaline, “feeling pumped up”) could actually help you do well, not worse	☐	☐	☐	☐	☐
Do something to take your mind off the test	☐	☐	☐	☐	☐
Ignore your anxiety: Focus on studying or on the test instead	☐	☐	☐	☐	☐
Spend a few moments writing about your worries	☐	☐	☐	☐	☐

If forced to choose, would you say that anxiety and physiological arousal, on average, helps performance or hurts performance?

- ☐ Helps performance
- ☐ Hurts performance

Are you majoring in Psychology?

- ☐ Yes
- ☐ No

Are you minoring in Psychology?

- ☐ Yes
- ☐ No

After the term in which you took [Intro Psych], how many additional psychology courses have you taken?

- ☐ 0
- ☐ 1
- ☐ 2
- ☐ 3
- ☐ 4
- ☐ 5
- ☐ 6
- ☐ 7
- ☐ 8
- ☐ 9
- ☐ 10 or more

What is your favorite psychology class that you have taken at [School]?

Why is this class your favorite psychology class?

We are interested in your knowledge of some of the content is covered in [Intro Psych]. So, the next part of the survey is a quick quiz (16 questions) of psychology content knowledge. Please answer the questions to the best of your ability WITHOUT using any external aids (e.g., notes, the internet, a friend). We want to get a sense of what content students remember from [Intro Psych], years later. If you don't know an answer, please give your best guess.

According to the [Intro Psych] textbook, which of the following is a theme that unites the field of psychology?

- ☐ Understanding complex psychological phenomena requires multiple perspectives.
- ☐ People are not passive: they are active in selecting, interpreting, and organizing their experience.
- ☐ Psychology involves trade-offs: the price we pay for processes that are mostly useful is that those same processes occasionally cause problems.
- ☐ All of the above are themes that unite psychology.

Which of the following statistics is used to determine how much variation there is in a sample, around the sample mean?

- ☐ mode
- ☐ median
- ☐ distribution function
- ☐ standard deviation

Amy is a health psychologist who wants to study whether low-impact exercise or high-impact exercise is more effective in helping people to maintain a healthy weight. She asks participants from a local gym whether they participate in predominantly low- or high-impact exercise and then computes their Body Mass Index. She then compares the two groups (low- and high-impact exercisers) to see if there are differences in how healthy their weight is. Which of the following statements is TRUE?

- ☐ Amy's independent variable is the Body Mass Index of the participants
- ☐ Amy's has randomly assigned her participants to conditions.
- ☐ Amy's study can be described as a quasi-experiment.
- ☐ All of the above are true.

Which of the following is TRUE of the use of deception in research on human participants?

- ☐ Deception can never be used because of ethical concerns.
- ☐ Deception is used in most psychological research.
- ☐ Deception should be minimized for ethical reasons and followed by careful debriefing.
- ☐ Deception is not a problem for psychological research because all participants are debriefed.

Genes interact with an organism's environment to influence the development of the organism's observable characteristics or traits, which are also called the organism's:

- ☐ genotype.
- ☐ phenotype.
- ☐ dominant gene.
- ☐ alleles.

According to the [Intro Psych] textbook, the fact that babies who are blind since birth begin to smile at the same time as babies with normal vision tells us that _____.

- ☐ smiling is a learned response
- ☐ smiling is species specific to humans
- ☐ smiling is species general in humans
- ☐ smiling occurs only if a baby is happy

According to lecture, which of Darwin's three "facts of life" did he and other scientists have the least understanding of (in terms of being able to explain the underlying mechanism) at the time that his theory was published?

- ☐ variation
- ☐ heredity
- ☐ competition
- ☐ survival of the fittest

The resting potential of a neuron

- ☐ is an electrical phenomenon unique to the giant squid neuron.
- ☐ refers to the electrical current associated with propagation of an action potential.
- ☐ refers to the fact that the inside of a neuron is negatively charged relative to the outside.
- ☐ depends on what kind of neuron it is.

According to the [Intro Psych] textbook, which of the following is a true limitation in the plasticity of the central nervous system (CNS)?

- ☐ Neurons in the CNS cannot develop new synapses.
- ☐ The CNS cannot grow new neurons to replace those damaged through injury.
- ☐ Neurons in the CNS cannot change how much neurotransmitter they release in the synapse.
- ☐ Neurons in the CNS cannot change how sensitive they are to neurotransmitters.

Which of the following brain regions is most involved in helping a person coordinate complex actions like taking a drink from their water bottle without spilling it?

- ☐ the cerebellum
- ☐ the pons
- ☐ the amygdala
- ☐ the hippocampus

In reversible figures like the one above, figure and ground relations are deliberately reversible and ambiguous. According to the [Intro Psych] textbook, such ambiguity illustrates which important general point about human perception?

- ☐ Adaptation to steady or repeated stimulation happens in almost all of the senses.
- ☐ Only a few basic perceptual abilities are inborn.
- ☐ Figure-ground relationships are inherent in the stimulus.
- ☐ The perceiver plays an active role in interpreting input.

You wake up in the middle of the night. You don't want to wake your roommate, so you try to find your shoes in the dark. Whatever you are able to see is thanks to what type of receptor on your retina?

- ☐ cones
- ☐ rods
- ☐ horizontal cells
- ☐ bipolar cells

Your roommate spends several hours one Saturday playing Guitar Hero, a video game that involves interpreting different lights that move in a constant downward direction. When you finally pry him away from his game, he reports that the entire room looks like it is moving upward. Based on the lecture on perception, you deduce that your roommate is experiencing:

- ☐ induced motion of the self
- ☐ apparent motion
- ☐ optic flow
- ☐ the motion aftereffect

You would like to train a very attractive classmate to salivate when he or she sees you. To do so, you offer your classmate a super sour candy every time you see him or her in class. After a few classes, sure enough, this classmate salivates as soon you arrive. In this example, what is the unconditioned stimulus (UCS) and what is the conditioned stimulus (CS)?

- ☐ you; the sour candy
- ☐ you; salivation
- ☐ the sour candy; you
- ☐ the sour candy; salivation

Which of the following forms of learning best illustrates that "not all associations are created equal." In other words, that animals (including humans) are biologically prepared to learn certain types of associations more easily than others?

- ☐ drug cravings
- ☐ conditioned emotional responses
- ☐ learned taste aversion (Garcia effect)
- ☐ learned helplessness

Evidence for observational learning was found in which of the following studies?

- ☐ Thorndike's cats-in-puzzle-box studies
- ☐ Watson's "Little Albert" studies
- ☐ Bandura's Bobo doll studies
- ☐ Skinner's pigeon studies

In the quiz, how many questions did you guess on?

- ☐ All or almost all
- ☐ More than half
- ☐ About half
- ☐ About a quarter
- ☐ One or two
- ☐ None

We are also interested in how [Intro Psych] may have influenced your experiences in other classes or in your life more generally. Please answer the following questions about what you learned in [Intro Psych] that have been helpful to you in other classes or in your life generally.

What, if anything, did you learn in [Intro Psych] that has been useful to you in your other classes? Please consider content or new knowledge, study skills, ways to interact with others, ways to plan your courses, or anything else you learned.

What, if anything, did you learn in [Intro Psych] that has been useful to you in your life generally? Please consider content or new knowledge, study skills, ways to interact with others, or anything else you learned.

If you could give advice to students just starting [Intro Psych], what advice would you give them?

Do you have any other feedback about [Intro Psych]?

We want to learn more about the academic trajectories of students who take [Intro Psych]. To that end, it would be helpful to obtain your transcript. Please know that all of your information will be kept strictly confidential. All identifying information will be removed, your grades will never be connected with you, and only group-level results will ever be reported. This information would be very helpful to us. However, you can decline if you would like. In that case, please just move on to the next page.

Authorization of Transcript Release

I authorize the release of my undergraduate transcript from the Registrar's Office of [School] University to [Study Investigators]. I understand that my transcript will be destroyed after information concerning my grades has been transcribed onto another document linked to my identity only through a confidential subject number. I understand that my grades may be used in statistical analyses that aggregate across groups of students, and that these aggregate statistics may be published in written and/or spoken form (but that any identifying information will be destroyed). I understand that my personal grades will not be published in written or spoken form. I give my electronic signature by providing my name and date below.

Name (first and last)

Today's date

My [School] University ID# is... (numeric 8-digit ID number)

Thank you! You're done with the survey. We appreciate your time and perspective. Your responses will help us improve [Intro Psych] for future students.

In the space below, please provide the email address to which we should send your Amazon gift card. (Allow up to 2 weeks for delivery.) Then, click the button to submit the survey.